Supplementary Material

An Interpretable Machine Learning Framework for Early- versus Late-Stage Classification of Breast Cancer from Gene Expression, with Co-expression Network Analysis

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Table S4: Differential expression of all 65 panel genes between early-stage (AJCC Stage I–II; n = 805) and late-stage (Stage III–IV; n = 270) primary tumours (n = 1,075; two-sided Mann–Whitney U test), ordered by p-value.

Means are on the log2 expression scale; Difference, mean late minus mean early; r, rank-biserial correlation (positive, higher in late stage); FDR, Benjamini–Hochberg adjusted p-value across the 65 genes. No gene reached FDR < 0.05. Values are rounded as in Table 3 of the main text (values that round to zero are shown as 0.00); unrounded values are provided in the accompanying spreadsheet.

Rank	Gene	Mean early	Mean late	Difference	r	p	FDR
1	*CD44*	0.52	0.28	−0.24	−0.14	< 0.001	0.057
2	*MAPK3*	0.07	0.20	+0.13	+0.12	0.005	0.10
3	*JAK2*	0.37	0.21	−0.16	−0.11	0.005	0.10
4	*TGFB1*	−0.40	−0.27	+0.13	+0.11	0.007	0.11
5	*PALB2*	0.36	0.27	−0.09	−0.10	0.010	0.11
6	*FLT3*	2.13	1.74	−0.38	−0.10	0.014	0.11
7	*ALDH1A1*	−1.89	−1.65	+0.24	+0.10	0.014	0.11
8	*AKT1*	0.18	0.28	+0.10	+0.10	0.015	0.11
9	*SOX9*	0.73	0.54	−0.20	−0.10	0.017	0.11
10	*SMAD2*	0.23	0.15	−0.08	−0.10	0.017	0.11
11	*TGFBR2*	−0.25	−0.11	+0.14	+0.10	0.019	0.11
12	*VEGFA*	−0.47	−0.63	−0.16	−0.09	0.021	0.11
13	*STAT3*	0.32	0.24	−0.08	−0.09	0.024	0.12
14	*FGFR1*	0.69	0.85	+0.16	+0.09	0.029	0.13
15	*SMAD4*	0.01	−0.05	−0.06	−0.09	0.031	0.13
16	*PDGFRA*	0.65	0.84	+0.19	+0.09	0.032	0.13
17	*NFKB1*	0.20	0.12	−0.08	−0.09	0.033	0.13
18	*RUNX1*	1.60	1.49	−0.11	−0.08	0.050	0.18
19	*CHEK2*	0.23	0.11	−0.11	−0.08	0.055	0.18
20	*FOS*	−0.55	−0.79	−0.24	−0.08	0.055	0.18
21	*NFKB2*	−0.05	−0.15	−0.10	−0.07	0.109	0.34
22	*SOX2*	−3.21	−2.85	+0.36	+0.06	0.126	0.37
23	*BRCA2*	0.76	0.60	−0.16	−0.06	0.138	0.39
24	*MTOR*	0.04	−0.02	−0.06	−0.06	0.158	0.43
25	*PIK3CA*	0.35	0.27	−0.08	−0.06	0.165	0.43
26	*PROM1*	1.19	0.83	−0.35	−0.06	0.171	0.43
27	*BRAF*	0.26	0.21	−0.05	−0.05	0.196	0.47
28	*GATA3*	5.93	6.37	+0.43	+0.05	0.210	0.49
29	*NANOG*	−0.06	−0.12	−0.06	−0.05	0.217	0.49
30	*SRC*	−0.38	−0.44	−0.06	−0.05	0.232	0.50
31	*ERBB2*	1.38	1.53	+0.15	+0.05	0.239	0.50
32	*ATM*	−0.04	−0.12	−0.08	−0.05	0.254	0.50
33	*CDK4*	−0.14	−0.11	+0.03	+0.05	0.256	0.50

Rank	Gene	Mean early	Mean late	Difference	r	p	FDR
34	*EGFR*	−1.97	−2.19	−0.22	−0.04	0.272	0.51
35	*REL*	0.72	0.61	−0.11	−0.04	0.278	0.51
36	*ARID1B*	0.17	0.12	−0.04	−0.04	0.282	0.51
37	*CREBBP*	0.21	0.15	−0.05	−0.04	0.292	0.51
38	*NOTCH1*	−0.65	−0.65	0.00	+0.04	0.308	0.52
39	*SMAD3*	−0.43	−0.35	+0.07	+0.04	0.312	0.52
40	*KIT*	0.62	0.49	−0.13	−0.04	0.343	0.56
41	*MYC*	−0.03	−0.10	−0.07	−0.04	0.359	0.57
42	*CDK6*	−0.88	−1.03	−0.15	−0.04	0.378	0.58
43	*CTNNB1*	0.21	0.16	−0.05	−0.04	0.383	0.58
44	*POU5F1*	−1.66	−1.89	−0.23	−0.03	0.400	0.59
45	*NCOR2*	−0.25	−0.24	+0.01	+0.03	0.443	0.64
46	*RB1*	−0.17	−0.10	+0.07	+0.03	0.508	0.72
47	*KRAS*	0.19	0.18	−0.01	+0.03	0.537	0.74
48	*NCOR1*	0.17	0.17	0.00	+0.02	0.597	0.81
49	*MDM2*	0.18	0.18	0.00	−0.02	0.609	0.81
50	*TP53*	0.21	0.21	−0.01	−0.02	0.625	0.81
51	*ARID1A*	0.25	0.24	−0.01	−0.02	0.689	0.87
52	*ABL1*	−0.10	−0.09	0.00	+0.01	0.723	0.87
53	*BRCA1*	0.26	0.27	+0.01	−0.01	0.733	0.87
54	*RUNX2*	0.85	0.91	+0.05	+0.01	0.740	0.87
55	*EP300*	0.14	0.13	−0.01	+0.01	0.753	0.87
56	*PTEN*	0.24	0.22	−0.01	−0.01	0.762	0.87
57	*FOXA1*	4.93	5.49	+0.56	+0.01	0.775	0.87
58	*JUN*	−0.31	−0.32	−0.01	+0.01	0.776	0.87
59	*AR*	3.22	3.45	+0.23	+0.01	0.788	0.87
60	*ESR1*	5.16	5.60	+0.44	+0.01	0.837	0.90
61	*CCND1*	0.96	0.96	0.00	−0.01	0.842	0.90
62	*CDK2*	0.17	0.15	−0.02	0.00	0.905	0.93
63	*HIF1A*	−0.71	−0.70	+0.01	0.00	0.909	0.93
64	*CDH1*	1.47	1.40	−0.06	0.00	0.913	0.93
65	*MAPK1*	−0.17	−0.18	−0.01	0.00	0.968	0.97